

# Lecture Material #5

## Information Security

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# Modular Arithmetic

If  $a$  and  $b$  are integers and  $m$  is a positive integer, then  $a$  is congruent to  $b$  modulo  $m$  if  $m$  divides  $a - b$ . We use the notation  $a \equiv b \pmod{m}$  to indicate that  $a$  is congruent to  $b$  modulo  $m$ . If  $a$  and  $b$  are not congruent modulo  $m$ , we write

$$a \not\equiv b \pmod{m}$$





# Modular Arithmetic

Let  $a$  and  $b$  be integers, and let  $m$  be a positive integer. Then  $a \equiv b \pmod{m}$  if and only if

$$a \bmod m = b \bmod m$$





# Modular Arithmetic

Determine whether 17 is congruent to 5 modulo 6  
and whether 24 and 14 are congruent modulo 6.

*Solution:* Because 6 divides  $17 - 5 = 12$ , we see that  $17 \equiv 5 \pmod{6}$ . However, because  $24 - 14 = 10$  is not divisible by 6, we see that  $24 \not\equiv 14 \pmod{6}$ . ◀





# Modular Arithmetic

Let  $m$  be a positive integer. The integers  $a$  and  $b$  are congruent modulo  $m$  if and only if there is an integer  $k$  such that  $a = b + km$





# Modular Arithmetic

Let  $m$  be a positive integer. If  $a \equiv b \pmod{m}$  and  $c \equiv d \pmod{m}$ , then

$$a + c \equiv b + d \pmod{m} \quad \text{and} \quad ac \equiv bd \pmod{m}.$$

$$7 \equiv 2 \pmod{5} \text{ and } 11 \equiv 1 \pmod{5},$$

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}.$$





# Modular Arithmetic (Greatest Common Divisor)

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Let  $a$  and  $b$  be integers, not both zero. The largest integer  $d$  such that  $d \mid a$  and  $d \mid b$  is called the greatest common divisor of  $a$  and  $b$ . The greatest common divisor of  $a$  and  $b$  is denoted by  $\gcd(a, b)$ .





# Modular Arithmetic (GCD Linear Combination)

If  $a$  and  $b$  are positive integers, then there exist integers  $s$  and  $t$  such that  $\gcd(a, b) = sa + tb$







# Modular Arithmetic (GCD Linear Combination)

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Find the greatest common divisor (GCD) of **414** and **662** using the Euclidean algorithm.

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41.$$

Hence,  $\gcd(414, 662) = 2$ , because 2 is the last nonzero remainder.





# Modular Arithmetic (GCD Linear Combination)

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$$\gcd(252, 198)$$

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18.$$





# Modular Arithmetic (GCD Linear Combination)

Express  $\gcd(252, 198) = 18$  as a linear combination of 252 and 198.

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18.$$

$$18 = 54 - 1 \cdot 36$$

$$36 = 198 - 3 \cdot 54$$

$$\begin{aligned} 18 &= 54 - 1 \cdot 36 \\ &= 54 - 1 \cdot (198 - 3 \cdot 54) \\ &= 4 \cdot 54 - 1 \cdot 198 \end{aligned}$$

$$54 = 252 - 1 \cdot 198$$

$$\begin{aligned} \gcd(252, 198) &= sa + tb \\ 18 &= 4 \cdot 252 + (-5) \cdot 198 \end{aligned}$$

$$18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$$





# Modular Arithmetic (GCD Linear Combination)

*Example 5.* Find  $\gcd(41, 12)$  and express it as a linear combination of 41 and 12.





# Modular Arithmetic (Practice Problems)

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- ❑ Find the greatest common divisor (GCD) of 116 and 2040 using the Euclidean algorithm.
- ❑ Find the greatest common divisor (GCD) of 3279 and 2073 using the Euclidean algorithm.





# Modular Arithmetic (Linear Congruence)

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A **congruence** of the form

$$ax \equiv b \pmod{m}$$

where **m** is a positive integer, **a** and **b** are integers, and **x** is a variable, is called a ***linear congruence***.

To find **all the integers x satisfying** this congruence





# Modular Arithmetic (Linear Congruence)

$$ax \equiv b \pmod{m}$$

**Theorem:** If  $a$  and  $m$  are relatively prime integers and  $m > 1$ , then an **inverse of  $a$  modulo  $m$**  exists.

**Proof:** Since  $\gcd(a, m) = 1$ , there are integers  $s$  and  $t$  such that

$$sa + tm = 1$$

This implies that  $sa + tm \equiv 1 \pmod{m}$

Since  $tm \equiv 0 \pmod{m}$ , it follows that  $sa \equiv 1 \pmod{m}$

Consequently,  **$s$  is the inverse of  $a$  modulo  $m$**





# Modular Arithmetic (Linear Congruence)

Solve the congruence equation  $33X \equiv 38 \pmod{280}$

$$sa + tm = 1$$

$$ax \equiv b \pmod{m}$$

Check  $\rightarrow \text{GCD}(33, 280)$

Since,  $\text{GCD}(33, 280) = 1$ , the equation has a unique solution

Considering Euclidean Algorithm:  $33X \equiv 1 \pmod{280}$

Linear combination of  $\text{GCD}(33, 280) = 1 \rightarrow$

$$(17) \quad 33 + (-2)280 = 1$$

$$s \quad a + t \quad m = 1$$







# Modular Arithmetic (Linear Congruence)

Solve the congruence equation  $33X \equiv 38 \pmod{280}$

$$(17) \quad \begin{matrix} 33 \\ s \end{matrix} + \begin{matrix} (-2)280 \\ t \end{matrix} = \begin{matrix} 1 \\ 1 \end{matrix} \qquad \qquad \qquad a x \equiv b \pmod{m}$$

$s b = 17 \times 38 = 646$  is a solution of the given equation

Dividing 646 by  $m=280 \rightarrow$  **Remainder = 86**

86 is a **unique solution** of between **0 to 279**

The **general solution** is  $86 + 280k$  with  $k \in \mathbb{Z}$





# Modular Arithmetic



Solve the congruence equation  $195X \equiv 23 \pmod{968}$

293



